

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			Any five (x1) from: A. Translation (1) B. tRNA brings an amino acid to the ribosome (1) C. Each tRNA has a {specific/ particular} amino acid (1) D. Complementary base pairing occurs between the tRNA anticodon and the mRNA codon/ description of codon anticodon interaction (1) E. A second amino acid is brought to the ribosome / two tRNA molecules can attach to the mRNA (at the ribosome)/ reference to two binding sites. (1) F. A condensation reaction occurs/ a peptide bond forms between the amino acids. (1) G. The ribosome moves along the mRNA {by one codon/until a stop codon is reached.} (1)	5			5		
	(d)			{Thymine/ T} is replaced with {adenine/ A} / CAC. (1)		1		1		
	(e)	(i)		Both R groups circled	1			1		
		(ii)		Phenylalanine, leucine and valine are all non- polar and therefore are attracted to each other (1) Glutamic acid is polar and will therefore {be repelled from/ not attracted to} phenylalanine and leucine (1) any correct reference to hydrophobic/ hydrophilic (1)			3	3		
				Question 4 total	9	6	3	18	1	0